RAJALAKSHMI ENGINEERING COLLEGE

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Y2

Started on	Wednesday, 30 September 2026, 9:38 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:39 AM
Time taken	36 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the duplicate element using Floyd's Cycle Detection
int findDuplicate(int arr[], int n) {
    // Phase 1: Finding the intersection point in the cycle
    int tortoise = arr[0];
    int hare = arr[0];

    do {
        tortoise = arr[tortoise];       // Moves 1 step
        hare = arr[arr[hare]];          // Moves 2 steps
    } while (tortoise != hare);

    // Phase 2: Finding the entrance to the cycle (the duplicate element)
    tortoise = arr[0];
    while (tortoise != hare) {
        tortoise = arr[tortoise];       // Moves 1 step
        hare = arr[hare];               // Moves 1 step
    }

    return tortoise;
}

int main() {
    int n;

    // Read the size of the array
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    // Dynamically allocate memory for the array
    int *arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) {
        return 1;
    }

    // Read the array elements
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Find and print the duplicate element
    int duplicate = findDuplicate(arr, n);
    printf("%d\n", duplicate);

    // Free memory
    free(arr);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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